

M2 Quantitative Finance
XVAs

Cost of Hedging XVA

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Abstract

This report provides a detailed analysis of the Hedging Valuation Adjustment (HVA) framework introduced by [Burnett \(2021\)](#) and extended by [Burnett and Williams \(2021\)](#), offering a unified treatment of hedging friction costs within the standard XVA architecture. We derive an explicit formulation of the HVA consistent with other valuation adjustments such as CVA and FVA, decomposing it into a drag component, capturing the continuous cost of delta rebalancing under bid–ask spreads, and a closeout component, reflecting discrete losses upon counterparty default.

We replicate the numerical results of the original framework using Monte Carlo simulations and provide a detailed sensitivity analysis. Our findings confirm that the HVA can be of the same order of magnitude as classical XVAs and may represent a non-negligible proportion of CVA, with credit hedging frictions identified as the primary driver, while asset-related and closeout components remain comparatively limited.

Finally, we extend the framework through a simplified empirical application calibrated to WTI crude oil market data. In this setting, we find that the resulting HVA is negligible, suggesting that hedging costs are highly dependent on both the modelling assumptions and the level of market frictions. This result highlights an important limitation of the HVA framework in practical applications, where realistic market conditions may significantly dampen its quantitative impact.

Keywords: XVA, HVA, CVA, transaction costs, hedging frictions, Monte Carlo simulation.

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1 Introduction

In modern **over-the-counter** (OTC) **derivatives markets**, **valuation adjustments** (XVAs) have become a central component of pricing and risk management, particularly since the financial crisis. These adjustments account for various **market imperfections** and **institutional costs** that are absent from a purely risk-free valuation framework. More precisely, XVAs correspond to **corrections** applied to the **theoretical price** of a derivative in order to incorporate risks and costs arising from the trading environment. In a standard replication-based pricing framework, the value of a derivative is obtained through a **hedging strategy**, and the initial premium corresponds to the cost of setting up this hedge. However, in practice, several additional factors must be taken into account, leading to the introduction of valuation adjustments.

XVAs represent the impact of **counterparty risk**, **funding constraints**, and **regulatory requirements** on the valuation of OTC derivatives. These adjustments are typically computed and charged to the client, and managed dynamically over time through dedicated accounts. As a consequence, XVAs are not static quantities, but rather **stochastic processes** whose variations generate additional profit and loss (P&L) for the bank.

The most common valuation adjustments include the **Credit Valuation Adjustment** (CVA), which reflects the expected loss due to counterparty default; the **Debt Valuation Adjustment** (DVA), which accounts for the bank's own default risk; the **Funding Valuation Adjustment** (FVA), capturing the cost of funding the hedging strategy; the **Margin Valuation Adjustment** (MVA), related to collateral requirements; and the **Capital Valuation Adjustment** (KVA), reflecting the cost of regulatory capital. Together, these components provide a more realistic valuation of derivatives in incomplete and frictional markets.

In this context, [Burnett \(2021\)](#) introduced the concept of **Hedging Valuation Adjustment** (HVA) as a way to **quantify** the **impact of hedging frictions** on the value of a derivatives portfolio, in a manner analogous to standard XVAs. However, this initial formulation did not address how HVA interacts with other, more classical valuation adjustments. Therefore, [Burnett and Williams \(2021\)](#) extend the original framework in order to derive a **formulation of HVA** that is **fully compatible with existing valuation adjustments**, allowing for a unified treatment of hedging costs alongside counterparty risk, funding and other market frictions.

More precisely, [Burnett and Williams \(2021\)](#) generalise the previous results in order to incorporate several sources of hedging-related costs. The resulting HVA captures friction effects associated with **hedging** both the **underlying asset** and **counterparty credit risk**, and also includes a contribution arising from the **unwind of hedges** at the counterparty's default. In this way, [Burnett and Williams](#) propose a more complete valuation adjustment that reflects the actual costs faced by institutions when dynamically managing XVA sensitivities. [Burnett and Williams](#) explain that the motivation for introducing such an adjustment is particularly strong for XVA desks. Unlike traditional trading desks, which may control their exposures through deal flow or hedge efficiently using liquid instruments, XVA desks are often exposed to **second-order** and **cross risks** at the portfolio level that cannot be perfectly hedged.

The goal of this report is to analyse and reproduce the main theoretical and numerical results presented in [Burnett and Williams \(2021\)](#). In particular, we aim to provide a clear description of the mathematical framework used to define the **Hedging Valuation Adjustment**, to explain how it is derived and how it interacts with other **XVAs**, and to highlight the role played by **hedging frictions** and **higher-order risks** in valuation.

In addition, we extend this framework through a simplified empirical application based on real market data (**WTI crude oil**), in order to assess the practical relevance of the **HVA**. This allows us to analyse how hedging costs behave in a more realistic setting and to evaluate the sensitivity of the **HVA** to key parameters such as volatility, transaction costs, hedging tolerance and maturity. Our results highlight the strong dependence of the **HVA** on modelling assumptions and market frictions, providing further insight into its practical significance.

1.1 Motivation and hypothesis

The increasing importance of valuation adjustments in derivatives pricing has highlighted the need to account for various sources of costs beyond classical risk-neutral valuation. While adjustments such as **CVA**, **DVA** or **FVA** are now well-established, the cost generated by hedging strategies in the presence of market frictions remains less systematically incorporated into pricing frameworks.

Our main motivation is to better understand and quantify the impact of these hedging frictions through the concept of the **Hedging Valuation Adjustment**, and to analyse how this adjustment interacts with standard **XVAs** in a consistent framework. This leads us to the following research questions:

- Q1** How can the **HVA** be rigorously defined within a framework consistent with classical **XVAs**?
- Q2** What are the main sources of hedging costs (**underlying asset hedging**, **counterparty credit hedging**, **hedge unwinds at default**), and how do they contribute quantitatively to the **HVA**?
- Q3** To what extent can the **HVA** be comparable in magnitude to standard **XVAs** under realistic market conditions?
- Q4** To what extent does the **HVA** remain relevant when applied to real market data under simplified but realistic assumptions?

A second motivation of this work is to bridge the gap between the theoretical framework proposed in the paper and its practical implementation. In particular, we aim to assess whether the mechanisms described by the authors can be reproduced numerically in a simplified setting, and to analyse the sensitivity of the **HVA** to key modelling assumptions.

More specifically, our empirical analysis suggests that while the **HVA** can be significant in stylised frameworks, its magnitude may become negligible in simplified real-market applications. This highlights the strong dependence of hedging costs on market frictions and modelling assumptions, and motivates a critical assessment of the practical relevance of the **HVA**.

1.2 Contributions

Our main contributions are the following.

First, we provide a detailed and structured presentation of the modelling framework introduced by [Burnett and Williams \(2021\)](#), with particular emphasis on its financial interpretation and its consistency within the broader **XVA** architecture.

Second, we implement a numerical version of the model and successfully replicate the key mechanisms and results highlighted in the original paper. We further extend this analysis by investigating the sensitivity of the **Hedging Valuation Adjustment (HVA)** to key parameters such as volatility, transaction costs, hedging tolerance and maturity, allowing for a deeper understanding of its quantitative behaviour.

Third, we introduce a simplified empirical application based on real market data (**WTI crude oil**), providing a practical perspective on the theoretical framework. This application enables us to assess the behaviour of hedging costs in a more realistic setting.

2 Model framework

In this section, we present the modelling framework introduced in [Burnett and Williams \(2021\)](#), which allows us to derive a formulation of the HVA consistent with the standard XVA setting. The aim is to construct a tractable yet representative setup that captures the main sources of hedging costs while remaining analytically manageable.

This framework extends the classical XVA methodology by accounting not only for **counterparty and funding risks**, but also for the **cost of dynamically hedging** the resulting exposures. While **traditional adjustments** such as CVA and FVA are primarily driven by **counterparty credit risk** and **funding conditions**, the HVA captures a fundamentally different effect, namely the **systematic cost induced by hedging frictions**.

The section is organised as follows. First, we introduce the **model setup**, including the decomposition of the portfolio value into clean value, standard XVA components, and hedging costs. Then, we describe the **dynamics of the hedging portfolio** and identify the different sources of risk. Next, we specify the **hedging strategy** used to eliminate diffusion and jump risks. Finally, we introduce a **continuous-time representation of hedging costs**, which allows us to derive a valuation equation where HVA appears as an explicit adjustment.

2.1 Model setup

The analysis is carried out within a **simplified XVA framework** that focuses on the most relevant adjustments: CVA and FVA, allowing to isolate the interaction between valuation adjustments and hedging frictions without introducing unnecessary complexity. Moreover, a key feature of the model is that it is formulated at the level of the **entire derivative portfolio**. Instead of considering each counterparty separately, the XVA desk is assumed to manage **aggregated exposures** across all counterparties and originating books. This reflects the practical organisation of financial institutions, where hedging decisions are typically made on a **net portfolio basis**.

Within this framework, each **counterparty** c contributes a **clean value** V_c , corresponding to the risk-free mark-to-market (MtM) of the position, as well as an **XVA component** U_c , which captures **counterparty risk** and **funding effects**. In addition, **each book** j , including the XVA desk itself, generates a **hedging cost** denoted by H_j , reflecting the presence of **market frictions** in dynamic hedging strategies.

Under these assumptions, we can define the **total value** $\hat{V}$ of the **multi-counterparty and multi-book portfolio**.

Definition 2.1. (Total portfolio value)

The **total value** of the **multi-counterparty** and **multi-book portfolio** is defined as:

$$\hat{V} = \sum_c (V_c + U_c) + \sum_j H_j, \quad (1)$$

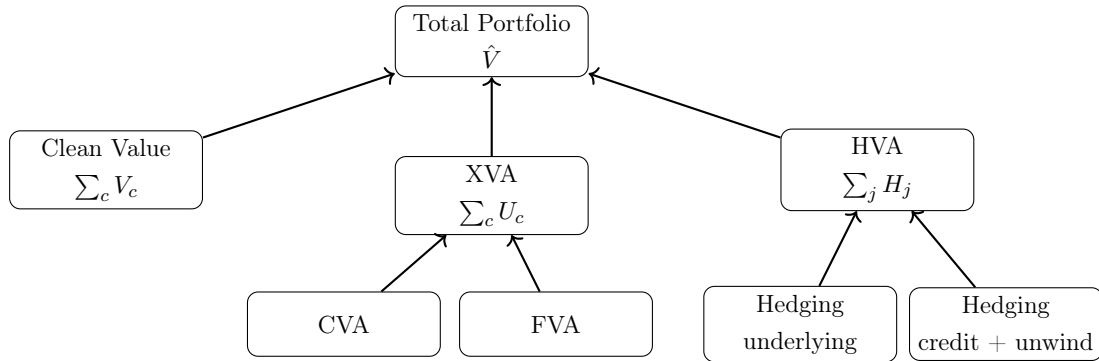
where the HVA of each book j H_j is:

$$H_j = \sum_c H_j^c, \quad (2)$$

with H_j^c the portion of the HVA of book j attributed to counterparty c .

This decomposition highlights that, in addition to standard valuation adjustments, the overall portfolio value must incorporate an **explicit cost** associated with **hedging**, which is captured by the **HVA term**. This structure, presented in Figure 1, reflects an important modelling distinction. Standard valuation adjustments, such as CVA and FVA, are attached to **counterparties**, whereas HVA is introduced at the **level of each book**. This separation is essential in order to identify HVA as a distinct adjustment rather than as a hidden component of FVA.

Figure 1: Decomposition of the portfolio value including XVA and HVA components.
Source: Own elaboration.



In addition, [Burnett and Williams \(2021\)](#) adopt several simplifying assumptions in order to keep the framework analytically tractable. First, **collateral is ignored**. As pointed out by [Burnett and Williams](#), collateral could in principle be incorporated into the setup, but this would mainly modify the classical CVA and FVA terms and would unnecessarily complicate the exposition at this stage. Second, the model assumes **deterministic interest rates**, **credit spreads** and **funding spreads**. Third, the **institution's own default is not hedged**. Finally, the **HVA itself is assumed to be hedged**, and this hedge is unwound when the relevant counterparty defaults.

These assumptions are clearly stylised, but they serve a precise purpose. The goal is not to provide the most general XVA model, but to derive in the simplest possible setting a **coherent formula for HVA** and to understand how it interacts with the existing valuation adjustments.

Next, we define the **hedge portfolio** associated with the derivative position. In line with replication-based XVA frameworks, the **total value of the portfolio** consists of positions in the **underlying asset**, **counterparty bonds**, and several **funding accounts**.

Definition 2.2. (Total hedging portfolio value)

The **total value of the hedging portfolio** is given by:

$$\Pi = \chi S + \sum_c \alpha_c P_c + \beta_S + \sum_c \beta_c + \sum_c \beta_c^F + \sum_j \tilde{\beta}_j^F + \hat{V}, \quad (3)$$

where each term represents a **specific component** of the **hedging strategy**. The position χ corresponds to **holdings in the underlying asset** S , while α_c denotes the **position** in the **zero-recovery bond** P_c of **counterparty** c . The remaining terms correspond to various **funding accounts** associated with the hedging strategy. More precisely, the **cash account** β_S is a **secured funding account** associated with the underlying position, accruing at the **repo rate** q_S **net of dividends** γ . For each counterparty c , the **account** β_c is used to finance the **position** in the corresponding **bond** P_c and accrues at the **repo rate** q_c . In addition, the **unsecured funding accounts** β_c^F are used to finance both the **clean mark-to-market** V_c and the **standard (non-HVA) valuation adjustments** U_c , and accrue at the **funding rate** r_F . Finally, the **accounts** $\tilde{\beta}_j^F$ are specifically dedicated to funding the HVA associated with **each book** j , also accruing at **rate** r_F .

The different **funding accounts** are linked to the portfolio positions through the following constraints:

$$\beta_S = -\chi S, \quad \beta_c = -\alpha_c P_c, \quad \beta_c^F = -(V_c + U_c), \quad \tilde{\beta}_j^F = -H_j. \quad (4)$$

These relations reflect the fact that **all positions** in the **hedge portfolio must be financed**. In particular, the last equality highlights a key feature of the model: the **HVA itself requires funding**, and therefore contributes directly to the overall cost of the hedging strategy.

The **dynamics** of the different components are given by:

$$dS = \mu(S, t) dt + \sigma(S, t) dW_t \quad (5)$$

$$dP_c = (q_c + \lambda_c) P_c dt - P_c dJ_c \quad (6)$$

$$d\beta_S = (q_S - \gamma) \beta_S dt = -(q_S - \gamma) \chi S dt \quad (7)$$

$$d\beta_c = q_c \beta_c dt = -q_c \alpha_c P_c dt \quad (8)$$

$$d\beta_c^F = r_F \beta_c^F dt = -r_F (V_c + U_c) dt \quad (9)$$

$$d\tilde{\beta}_j^F = r_F \tilde{\beta}_j^F dt = -r_F H_j dt \quad (10)$$

Here, each process J_c represents the default of counterparty c and is modelled as a **Poisson jump process**. In addition, **defaults** are assumed to be **independent across counterparties** and **independent** of the **market Brownian Motion (BM)**.

2.2 Portfolio dynamics

We consider the total value of the portfolio given in (3) together with the dynamics of its individual components specified in (5) - (10). Then, to characterise the **evolution** of the **total value of the hedging potfolio**, we compute the differential of each component and aggregate the resulting contributions.

Proposition 2.1. (Unhedged portfolio dynamics)

The infinitesimal variation of the hedging portfolio is given by:

$$d\Pi = \left(\hat{V}_t + \frac{1}{2}\sigma^2 \hat{V}_{SS} + \sum_c \alpha_c \lambda_c P_c - (q_S - \gamma)\chi S - r_F \hat{V} \right) dt + (\chi + \hat{V}_S) dS + \sum_c (\Delta \hat{V}_c - \alpha_c P_c) dJ_c - d\phi. \quad (11)$$

Each term in (11) has a precise interpretation. The quantity $\hat{V}_t$ is the **time derivative** of the adjusted portfolio value, while $\frac{1}{2}\sigma^2 \hat{V}_{SS}$ is the **gamma term** arising from the diffusion of the underlying asset. The term $\sum_c \alpha_c P_c \lambda_c$ represents the **drift contribution** of the counterparty bonds, whose expected return is linked to their **default intensity** λ_c . The quantity $(q_S - \gamma)\chi S$ corresponds to the **net carry of the position** in the underlying, where q_S denotes the financing or repo rate of the asset and γ its **dividend yield**. The term $r_F \hat{V}$ is the **funding cost** applied to the derivative position. Finally, $d\phi$ denotes the **infinitesimal friction cost** generated by the hedging strategy.

From a risk-management perspective, the dynamics can be decomposed into their main sources of risk as follows:

(i) **Diffusion exposure.** The term multiplying dS ,

$$(\chi + \hat{V}_S) dS, \quad (12)$$

represents the **sensitivity** of the **total portfolio** to **infinitesimal fluctuations** in the **underlying asset**. Therefore, it captures the **diffusion risk** associated with market movements.

(ii) **Jump exposure.** The jump term,

$$\sum_c (\Delta \hat{V}_c - \alpha_c P_c) dJ_c, \quad (13)$$

represents the **residual exposure** to **counterparty default**, capturing the **jump-to-default risk** that must be hedged using positions in counterparty bonds. Here, dJ_c denotes the default indicator process of counterparty c , and $\Delta \hat{V}_c$ is the jump in the derivative value upon default.

Proof. First, applying *Itô's lemma* to $\hat{V} = \hat{V}(t, S)$ yields:

$$d\hat{V} = \hat{V}_t dt + \hat{V}_S dS + \frac{1}{2}\sigma^2 \hat{V}_{SS} dt + \sum_c \Delta \hat{V}_c dJ_c, \quad (14)$$

where $\Delta \hat{V}_c$ represents the jump in portfolio value upon default of counterparty c .

Consequently, the **total portfolio variation** given by expression (11) can be rewritten as:

$$d\Pi = \chi dS + \sum_c \alpha_c dP_c + d\beta_S + \sum_c d\beta_c + \sum_c d\beta_c^F + \sum_j d\tilde{\beta}_j^F + d\hat{V} - d\phi. \quad (15)$$

Finally, substituting the expressions (5) - (10) and (14) into (15) and simplifying, we obtain:

$$\begin{aligned}
d\Pi &= \chi dS + \sum_c \alpha_c \{ (q_c + \lambda_c) P_c dt - P_c dJ_c \} - (q_S - \gamma) \chi S dt - \sum_c q_c \alpha_c P_c dt + \\
&\quad - \sum_c r_F (V_c + U_c) dt - \sum_j r_F H_j dt + \hat{V}_t dt + \hat{V}_S dS + \frac{1}{2} \sigma^2 \hat{V}_{SS} dt + \sum_c \Delta \hat{V}_c dJ_c - d\phi \\
&= \left(\hat{V}_t + \frac{1}{2} \sigma^2 \hat{V}_{SS} + \sum_c \alpha_c \lambda_c P_c - (q_S - \gamma) \chi S - r_F \hat{V} \right) dt + (\chi + \hat{V}_S) dS \\
&\quad + \sum_c (\Delta \hat{V}_c - \alpha_c P_c) dJ_c - d\phi.
\end{aligned}$$

□

2.3 Hedging strategy

Now, we derive the **infinitesimal change** of the **XVA-adjusted hedging portfolio** and explain how the hedging conditions are chosen. The goal is to characterise the evolution of the total portfolio value Π over an infinitesimal time interval dt . In particular, we aim to express $d\Pi$ in terms of its fundamental sources of risk: (1) **market risk** driven by the underlying asset, (2) **jump risk** associated with counterparty defaults, and (3) **deterministic drift terms** arising from funding and carry effects.

We specify the hedging strategy by eliminating the main sources of risk identified in the portfolio dynamics. In particular, we aim to neutralise both the **diffusion risk** driven by the underlying asset and the **jump risk** associated with counterparty defaults. The hedging strategy therefore consists of two main components: (i) **delta hedging** of the underlying asset, and (ii) **hedging of jump-to-default risk** through positions in counterparty bonds.

Proposition 2.2. (Hedged portfolio dynamics)

The portfolio dynamics after imposing the hedging conditions reduce to:

$$d\Pi = \left(\hat{V}_t + (q_S - \gamma) S \hat{V}_S + \frac{1}{2} \sigma^2 S^2 \hat{V}_{SS} - r_F \hat{V} + \sum_c \lambda_c (\Delta \hat{V}_c - \Psi_c) \right) dt - d\phi_{\text{drag}}. \quad (16)$$

Proof. We start from the full dynamics of the hedging portfolio given in (11) and impose the hedging conditions in order to eliminate the main sources of risk.

- (i) **Diffusion risk.** To eliminate the **exposure** associated with the **underlying asset**, we impose the standard **delta-hedging** condition:

$$\chi = -\hat{V}_S. \quad (17)$$

With this choice, the coefficient of dS in expression (11) vanishes.

- (ii) **Jump risk.** A subtle but important point is that a counterparty default does not only generate a jump in the derivative value, it also forces the hedging strategy to be unwound or restructured,

thereby creating **additional transaction costs**. For this reason, the friction term $d\phi$ itself may contain jump components proportional to the default indicators dJ_c .

The term $d\phi$ represents the **cumulative transaction costs** associated with **dynamic hedging**. These costs arise from two distinct mechanisms. First, **continuous rebalancing of the hedge** in response to market movements generates a systematic cost over time. Second, upon the **default of a counterparty**, the hedge must be **unwound**, leading to an additional discrete cost. To capture these effects, we decompose the friction term as:

$$d\phi = d\phi_{\text{drag}} + \sum_c \Psi_c dJ_c, \quad (18)$$

where $d\phi_{\text{drag}}$ denotes the **continuous hedging cost** (or frictional drag), and Ψ_c represents the **additional cost incurred** when unwinding hedges at the default of counterparty c .

The jump part if we substitute (18) into (11) becomes:

$$\sum_c \left(\Delta \hat{V}_c - \alpha_c P_c - \Psi_c \right) dJ_c. \quad (19)$$

Then, to eliminate the **jump exposure** associated with each counterparty default, we choose the bond position α_c so that the coefficient of dJ_c vanishes. This gives the condition:

$$\alpha_c P_c = \Delta \hat{V}_c - \Psi_c. \quad (20)$$

This relation has a clear interpretation: the counterparty bond position must replicate not only the jump in the derivative value at default, $\Delta \hat{V}_c$, but also the extra friction cost Ψ_c that is triggered by the default event.

Applying the hedging conditions (17) and (20) to infinitesimal variation of the hedging portfolio (11), both the diffusion term and the jump terms disappear and the portfolio dynamics reduce to:

$$d\Pi = \left(\hat{V}_t + (q_S - \gamma) S \hat{V}_S + \frac{1}{2} \sigma^2 S^2 \hat{V}_{SS} - r_F \hat{V} + \sum_c \lambda_c (\Delta \hat{V}_c - \Psi_c) \right) dt - d\phi_{\text{drag}}.$$

□

2.4 Continuous-time representation of hedging costs

The HVA is defined as the **expected cumulative cost of maintaining a hedging strategy**. In practice, hedging is performed at **discrete times**, and each rebalancing generates **transaction costs**, leading to a **systematic negative drift** in the portfolio's P&L. In order to isolate and quantify this effect, it is convenient to adopt a **continuous-time representation** of the **hedging framework**. While the original formulation considers discrete re-hedging intervals, we work in the limit where the **time step** becomes **infinitesimal**. This allows us to focus directly on the expected cost component of the P&L, while ignoring hedging errors that average out.

In this setting, **transaction costs** are modeled as a **continuous cash outflow** at a constant *friction rate*. Although this approach implicitly assumes that the portfolio is **perfectly hedged at all times**, it remains valid for valuation purposes.

Indeed, two distinct **P&L effects** arise from real-world hedging:

- (i) A **P&L variance component**, which results from imperfect hedging between rebalancing dates. Since the hedge is only updated discretely, the portfolio is temporarily exposed to market fluctuations, generating random gains and losses. This component has zero mean and therefore does not contribute to valuation in expectation.
- (ii) A **systematic P&L drift**, induced by transaction costs incurred when adjusting the hedge. Each rebalancing operation generates a cost, which accumulates over time and leads to a persistent negative drift in the portfolio's P&L.

The HVA captures the second effect only, while the first averages out in expectation. This fact leads to a modification of the classical pricing equation. In the presence of hedging frictions, the **self-financing condition** must be adjusted to account for the **continuous outflow** generated by **transaction costs**. The resulting dynamics of the hedging portfolio are characterised as follows.

Proposition 2.3. (Pricing equation with hedging frictions)

Under the self-financing condition in expectation, the **adjusted portfolio value** satisfies:

$$\mathbb{E}_t[d\Pi] = \left(\hat{V}_t + rx\hat{V}_x + \frac{1}{2}\sigma^2\hat{V}_{xx} - r\hat{V} \right) dt - \mathbb{E}_t[d\phi] = 0, \quad (21)$$

where $\hat{V} = \hat{V}(t, x)$ is the **adjusted value of the derivative position** (including valuation adjustments such as HVA); r is the **risk-free interest rate**; σ is the **volatility of the underlying asset**; and $d\phi$ denotes the infinitesimal *smoothed* friction cost incurred by the hedging strategy over the time interval dt .

Proof. Starting from the full specification obtained in (11), the dynamics of the portfolio are:

$$d\Pi = \left(\hat{V}_t + \frac{1}{2}\sigma^2\hat{V}_{xx} + \sum_c \alpha_c \lambda_c P_c - (q_x - \gamma)\chi x - r_F \hat{V} \right) dt + (\chi + \hat{V}_x)dx + \sum_c (\Delta \hat{V}_c - \alpha_c P_c) dJ_c - d\phi.$$

As we have seen in the previous point, the portfolio is subject to three sources of risk: **diffusion risk** driven by the underlying asset, **jump risk** associated with counterparty defaults, and **deterministic drift terms**. The hedging strategy is chosen so as to eliminate the stochastic sources of risk.

- (i) **Delta hedging.** The exposure to the underlying asset is removed by imposing the delta-hedging condition:

$$\chi = -\hat{V}_x. \quad (22)$$

Under this choice, the diffusion term vanishes:

$$(\chi + \hat{V}_x)dx = 0. \quad (23)$$

- (ii) **Jump hedging.** The exposure to counterparty default is eliminated by choosing the position in the counterparty bond such that:

$$\alpha_c P_c = \Delta \hat{V}_c. \quad (24)$$

This implies that the jump term disappears:

$$(\Delta \hat{V}_c - \alpha_c P_c) dJ_c = 0. \quad (25)$$

After imposing the hedging conditions (23) and (25), the portfolio dynamics reduce to a purely deterministic drift:

$$d\Pi = \left(\hat{V}_t + \frac{1}{2} \sigma^2 \hat{V}_{xx} + \sum_c \lambda_c \Delta \hat{V}_c - (q_x - \gamma) x \hat{V}_x - r_F \hat{V} \right) dt - d\phi. \quad (26)$$

At this stage, the expression still reflects the **full XVA framework**, where repo rates, funding costs, and default intensities are explicitly modelled.

In order to obtain a **tractable continuous-time formulation**, it is convenient to work with a simplified representation in which these effects are **absorbed into an effective drift operator**. In particular, we assume that **repo rates** and **funding rates coincide** with the **risk-free rate**, and that **dividend effects** are **negligible**. Under these assumptions, we replace:

$$(q_x - \gamma) x \hat{V}_x \approx r x \hat{V}_x, \quad r_F \approx r, \quad (27)$$

and neglect default-related drift terms.

This leads to the simplified dynamics:

$$d\Pi = \left(\hat{V}_t + r x \hat{V}_x + \frac{1}{2} \sigma^2 \hat{V}_{xx} - r \hat{V} \right) dt - d\phi, \quad (28)$$

and taking conditional expectations and using $\mathbb{E}_t[dW_t] = 0$, we obtain:

$$\mathbb{E}_t[d\Pi] = \left(\hat{V}_t + r x \hat{V}_x + \frac{1}{2} \sigma^2 \hat{V}_{xx} - r \hat{V} \right) dt - \mathbb{E}_t[d\phi]. \quad (29)$$

Finally, by imposing the self-financing condition we prove the proposition. $\square$

3 Closeout

In this section, we formalise the **impact of a counterparty default** on the **value of the hedged portfolio**. To this end, we refine the description of the portfolio dynamics by explicitly **incorporating** both **continuous hedging costs** and **discrete losses** arising at **default**.

The **closeout** mechanism describes the procedure by which a **derivative position** is **terminated** and **valued** at the **moment a counterparty defaults**. Economically, it reflects the **net amount** that is **effectively realised** once the **exposure** to the **defaulting counterparty** is **removed**. This includes: (1) the **recovery of outstanding claims**, (2) the **cancellation of future contractual obligations**, and (3) the **incorporation of valuation adjustments** such as XVA and HVA. From a modelling perspective, **default events** introduce **jump discontinuities** in the portfolio value, which must be **properly accounted for** to ensure **consistency** in replication arguments and valuation.

We start by defining the **closeout value** g_c , which represents the amount received at the default of counterparty c . By construction, this corresponds to the **recovered value of positive exposures minus any liabilities** owed to the counterparty. Formally, it is given by the following definition.

Definition 3.1. (Closeout value)

The **closeout value** represents the net amount realised upon the default of counterparty c , after accounting for recovery on positive exposures and full repayment of liabilities. It is defined as:

$$g_c = R_c V_c^+ - V_c^-, \quad (30)$$

where R_c denotes the **recovery rate** of counterparty c , $V_c^+ = \max(V_c, 0)$ is the **positive exposure** (i.e. the amount owed to the institution), and $V_c^- = \min(V_c, 0)$ is the **negative exposure** (i.e. the amount owed by the institution).

Equivalently, using the decomposition $V_c = V_c^+ + V_c^-$, the closeout value can be rewritten as:

$$g_c = R_c V_c^+ - (V_c - V_c^+) = V_c^+ (1 + R_c) - V_c. \quad (31)$$

This alternative expression will be convenient when analysing the jump in the portfolio value.

We now determine the **total jump in the portfolio value**, denoted by $\Delta \hat{V}_c$, occurring at the default of counterparty c . This jump captures three distinct effects:

- (i) the **removal** of the **risk-free mark-to-market value** V_c ,
- (ii) the **receipt** of the **closeout amount** g_c ,
- (iii) the **impact on the HVA**, which is non-linear at the portfolio level and cannot be decomposed additively across counterparties.

Definition 3.2. (Total jump at default)

The **total jump in the portfolio value** at the default of counterparty c is defined as the net change in the adjusted portfolio value resulting from the closeout of the position and the associated valuation adjustments. It is given by:

$$\Delta \hat{V}_c = -V_c + g_c - U_c - \sum_j m_j^c H_j^c, \quad (32)$$

where V_c is the **risk-free mark-to-market** of the position with counterparty c , g_c is the **close-out value**, U_c represents the **counterparty-specific valuation adjustments**, and $\sum_j m_j^c H_j^c$ denotes the **marginal contribution of counterparty c** to the portfolio-level HVA.

The last term deserves particular attention. Unlike standard XVAs, the **HVA** is defined at the **portfolio level** and **does not decompose linearly** across counterparties. Therefore, the impact of removing counterparty c must be captured through a marginal contribution, represented by a coefficient m_j^c , which measures the sensitivity of each book-level HVA component H_j to the removal of counterparty c .

Proposition 3.2. (Hedged portfolio dynamics with explicit closeout)

After incorporating the explicit expression of the jump at default, the hedged portfolio dynamics reduce to:

$$d\Pi = \left(\hat{V}_t + (q_S - \gamma)S\hat{V}_S + \frac{1}{2}\sigma^2 S^2 \hat{V}_{SS} - r_F \hat{V} - \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) \right) dt - d\phi_{\text{drag}}. \quad (33)$$

Proof. Substituting the expression of the closeout value (30) into the total jump equation (32), and using the decomposition $V_c = V_c^+ - V_c^-$, we obtain:

$$\begin{aligned} \Delta \hat{V}_c &= -V_c + V_c^+(1 + R_c) - V_c - U_c - \sum_j m_j^c H_j^c \\ &= -2(V_c^+ + V_c^-) + V_c^+ + V_c^+ R_c - U_c - \sum_j m_j^c H_j^c \\ &= -(1 - R_c)V_c^+ - 2V_c - U_c - \sum_j m_j^c H_j^c, \end{aligned} \quad (34)$$

Since negative exposures are settled at par at default, the term involving V_c^- does not contribute to the loss. Therefore, we obtain:

$$\Delta \hat{V}_c = -(1 - R_c)V_c^+ - U_c - \sum_j m_j^c H_j^c. \quad (35)$$

Finally, we insert the representation (35) of $\Delta \hat{V}_c$ into the hedged portfolio dynamics (16). $\square$

Following [Burnett and Williams \(2021\)](#), we define the HVA as the **quantity required to make the expected incremental P&L of the hedged portfolio equal to zero** under the real-world measure.

That is, $\mathbb{E}_t[d\Pi] = 0$. So, we now characterise the equation satisfied by the valuation adjustments by imposing the zero expected P&L condition.

Proposition 3.3. (XVA–HVA pricing equation)

Under the condition $\mathbb{E}_t[d\Pi] = 0$, the valuation adjustments U_c and H_j^c satisfy:

$$\sum_c \left(\partial_t + \mathcal{A}_t - (r_F + \lambda_c) \right) U_c + \sum_{j,c} \left(\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c) \right) H_j^c = (r_F - r_o)V + \sum_c \lambda_c (1 - R_c) V_c^+ + \sum_c \lambda_c \Psi_c + f(t), \quad (36)$$

where $\mathcal{A}_t$ is the **infinitesimal generator**.

This equation characterises the joint dynamics of the **standard valuation adjustments** U_c and the **hedging valuation adjustments** H_j^c . In particular, it shows that the HVA arises as the component required to compensate for hedging frictions $f(t)$, default losses $(1 - R_c)V_c^+$, and unwind costs Ψ_c , ensuring that the hedged portfolio has zero expected P&L under the real-world measure.

Proof. First, taking conditional expectations in (33), and noting that all drift terms are deterministic given the information at time t , we obtain:

$$0 = \left[\hat{V}_t + (q_S - \gamma)S\hat{V}_S + \frac{1}{2}\sigma^2 S^2 \hat{V}_{SS} - r_F \hat{V} - \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) \right] dt - \mathbb{E}_t[d\phi_{\text{drag}}]. \quad (37)$$

To express the contribution of hedging frictions in rate form, we introduce the *friction rate*, defined as the **average rate of loss due to day-to-day hedging frictions**:

$$f(t) = \frac{\mathbb{E}_t[d\phi_{\text{drag}}]}{dt}. \quad (38)$$

Dividing the previous equation by dt , we obtain:

$$\hat{V}_t + (q_S - \gamma)S\hat{V}_S + \frac{1}{2}\sigma^2 S^2 \hat{V}_{SS} - r_F \hat{V} = \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) - f(t). \quad (39)$$

We now decompose the adjusted value as:

$$\hat{V} = V + \sum_c U_c + \sum_j H_j, \quad (40)$$

where V is the clean value, U_c the standard XVAs, and H_j the HVA components.

Let us assume that the clean value satisfies the classical pricing equation:

$$V_t + (q_S - \gamma)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - r_o V = 0. \quad (41)$$

Then, we subtract the PDE (41) from the left-hand side of equation (39):

$$(\hat{V}_t - V_t) + (q_S - \gamma)S(\hat{V}_S - V_S) + \frac{1}{2}\sigma^2 S^2(\hat{V}_{SS} - V_{SS}) - (r_F \hat{V} - r_o V). \quad (42)$$

Using the decomposition defined in (40), we expand the derivatives as follows:

$$\hat{V}_t = V_t + \sum_c (U_c)_t + \sum_{j,c} (H_j^c)_t, \quad (43)$$

$$\hat{V}_S = V_S + \sum_c (U_c)_S + \sum_{j,c} (H_j^c)_S, \quad (44)$$

$$\hat{V}_{SS} = V_{SS} + \sum_c (U_c)_{SS} + \sum_{j,c} (H_j^c)_{SS}. \quad (45)$$

Substituting expressions (43) - (45) into (39):

$$\begin{aligned} & V_t + \sum_c (U_c)_t + \sum_{j,c} (H_j^c)_t + (q_S - \gamma)S \left(V_S + \sum_c (U_c)_S + \sum_{j,c} (H_j^c)_S \right) \\ & + \frac{1}{2}\sigma^2 S^2 \left(V_{SS} + \sum_c (U_c)_{SS} + \sum_{j,c} (H_j^c)_{SS} \right) - r_F \left(V + \sum_c U_c + \sum_{j,c} H_j^c \right) \\ & = \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) + f(t). \end{aligned} \quad (46)$$

Then, we separate the terms involving the clean value V from those involving the valuation adjustments:

$$\begin{aligned} & \left(V_t + (q_S - \gamma)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - r_F V \right) + \sum_c \left((U_c)_t + (q_S - \gamma)S(U_c)_S + \frac{1}{2}\sigma^2 S^2 (U_c)_{SS} - r_F U_c \right) \\ & + \sum_{j,c} \left((H_j^c)_t + (q_S - \gamma)S(H_j^c)_S + \frac{1}{2}\sigma^2 S^2 (H_j^c)_{SS} - r_F H_j^c \right) \\ & = \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) + f(t). \end{aligned} \quad (47)$$

Next, using the PDE satisfied by the clean value V ,

$$V_t + (q_S - \gamma)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - r_o V = 0, \quad (48)$$

we rewrite the first bracket in (47) as:

$$V_t + (q_S - \gamma)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - r_F V = -(r_F - r_o)V. \quad (49)$$

Substituting this identity into (47) gives:

$$\begin{aligned}
& \sum_c \left((U_c)_t + (q_S - \gamma)S(U_c)_S + \frac{1}{2}\sigma^2 S^2(U_c)_{SS} - r_F U_c \right) \\
& + \sum_{j,c} \left((H_j^c)_t + (q_S - \gamma)S(H_j^c)_S + \frac{1}{2}\sigma^2 S^2(H_j^c)_{SS} - r_F H_j^c \right) \\
& = (r_F - r_o)V + \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) + f(t).
\end{aligned} \tag{50}$$

Then, we distribute the default-intensity terms and move them inside the corresponding sums:

$$\begin{aligned}
& \sum_c \left((U_c)_t + (q_S - \gamma)S(U_c)_S + \frac{1}{2}\sigma^2 S^2(U_c)_{SS} - (r_F + \lambda_c)U_c \right) \\
& + \sum_{j,c} \left((H_j^c)_t + (q_S - \gamma)S(H_j^c)_S + \frac{1}{2}\sigma^2 S^2(H_j^c)_{SS} - (r_F + m_j^c \lambda_c)H_j^c \right) \\
& = (r_F - r_o)V + \sum_c \lambda_c (1 - R_c)V_c^+ + \sum_c \lambda_c \Psi_c + f(t).
\end{aligned} \tag{51}$$

Finally, introducing the infinitesimal generator:

$$\mathcal{A}_t = (q_S - \gamma)S\partial_S + \frac{1}{2}\sigma^2 S^2 \partial_{SS}, \tag{52}$$

we can write the previous equation in compact form as:

$$\begin{aligned}
& \sum_c \left(\partial_t + \mathcal{A}_t - (r_F + \lambda_c) \right) U_c + \sum_{j,c} \left(\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c) \right) H_j^c = \\
& (r_F - r_o)V + \sum_c \lambda_c (1 - R_c)V_c^+ + \sum_c \lambda_c \Psi_c + f(t).
\end{aligned}$$

□

4 Distilling to HVA

In this section, we aim to “**distill**” or **isolate** the component of the valuation equation that is specifically associated with hedging costs, i.e. to the HVA.

The derived equation (36) for the adjusted portfolio value includes both the **standard valuation adjustments** U_c , which account for counterparty credit risk and funding effects, and the additional terms H_j^c , which capture the **impact of hedging frictions**. In order to better understand the **role of HVA**, it is convenient to separate these different contributions.

First, we identify the part of the equation corresponding to the **classical adjustments** U_c .

Proposition 4.1. (Equation for standard XVA components)

The standard valuation adjustments U_c satisfy:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + \lambda_c)) U_c = (r_F - r_o)V + \sum_c \lambda_c (1 - R_c) V_c^+. \quad (53)$$

Then, we introduce a **discount factor** associated with the effective rate $r_F + \lambda_c$:

$$D_{r_F + \lambda_c}(t, u) = \exp \left(- \int_t^u (r_F + \lambda_c) ds \right). \quad (54)$$

Using the *Feynman–Kac representation*, we can express U_c as a discounted expectation under the risk-neutral measure.

Proposition 4.2. (Feynman–Kac representation of U_c)

The valuation adjustments admit the Feynman–Kac representation:

$$U_c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + \lambda_c}(t, u) (r_F - r_o) V_c(u) \right] du \quad (55)$$

$$- \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + \lambda_c}(t, u) \lambda_c (1 - R_c) V_c^+(u) \right] du. \quad (56)$$

The first term corresponds to the **FVA**, accounting for the differential between the funding rate r_F and the originator’s discount rate r_o . The second term corresponds to the **CVA**, arising from the expected credit loss on positive exposure V_c^+ , weighted by the counterparty’s hazard rate λ_c and loss-given-default $(1 - R_c)$.

After having characterised the standard XVA components U_c , we can now return to the **full valuation equation** (36). By subtracting equation (53) from (36), we eliminate the contributions related to CVA and FVA, thereby **isolating the terms associated with hedging costs**. This allows us to obtain an equation governing only the HVA components.

Proposition 4.3. (HVA dynamics)

The HVA components H_j^c satisfy:

$$\sum_{j,c} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = f(t) + \sum_c \lambda_c \Psi_c. \quad (57)$$

The left-hand side represents the **evolution of the HVA components** under the **combined effect of time, market risk** (through the diffusion operator $\mathcal{A}_t$), **funding**, and **counterparty default risk**. The right-hand side contains the **sources of hedging costs**. The first term, $f(t)$, corresponds to the expected continuous loss generated by the rebalancing of the hedge. The second term, $\sum_c \lambda_c \Psi_c$, accounts for the additional costs incurred when hedging positions must be unwound upon default.

In order to gain further insight, it is useful to decompose the **day-to-day drag friction bleed** $f(t)$ into its main components. In practice, hedging costs arise from two different sources: **trading** in the **underlying asset** and **trading in credit instruments** used to hedge counterparty risk.

Definition 4.1. (Decomposition of hedging frictions)

The friction rate $f(t)$ can be decomposed as:

$$f(t) = \sum_j f_j^S(t) + \sum_c f_c^C(t), \quad (58)$$

where f_j^S represents the **friction associated with hedging the exposure to the underlying asset**, and f_c^C corresponds to the **friction arising from hedging counterparty credit risk**.

Substituting this decomposition into equation (57), we obtain a more explicit representation of the **HVA dynamics**:

$$\sum_{j,c} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = \sum_j f_j^S + \sum_c f_c^C + \sum_c \lambda_c \Psi_c. \quad (59)$$

It is important to note that **not all desks contribute to the HVA in the same way**. In particular, the XVA desk typically manages **aggregated exposures** across multiple counterparties and products, leading to a different risk profile compared to standard trading desks. To make this distinction explicit, we split the index j into two categories: $j \neq X$, **originating desks** (non-XVA) and $j = X$, the **XVA desk**. Using this decomposition, the HVA equation can be rewritten as follows.

Proposition 4.4. (Distilled HVA equation)

The HVA dynamics can be written as:

$$\begin{aligned} & \sum_{c, j \neq X} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c + \sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_X^c \\ &= \sum_{j \neq X} f_j^S + f_X^S + \sum_c f_c^C + \sum_c \lambda_c \Psi_c. \end{aligned} \quad (60)$$

This decomposition highlights that the HVA is **not** a **homogeneous adjustment**, but rather the **aggregation of contributions** from **different desks**, each subject to its **own hedging strategy**, **funding conditions**, and **sources of friction**. In particular, the XVA desk plays a special role, as its exposure is typically concentrated and driven by **residual risks** that **cannot be fully hedged** at the level of **individual trades**. As a result, its contribution to the HVA may exhibit different sensitivities, as reflected by the multiplier m_X^c . Overall, this formulation makes clear that the total HVA emerges as the combined effect of hedging frictions across assets and counterparties, together with the additional costs generated at default through the unwind of hedging positions.

4.1 Originating desk HVAs

We now refine the analysis by distinguishing the contribution of the originating desks from that of the XVA desk. In particular, we **decompose the total HVA** into components associated with each desk and each counterparty.

Now, we focus on the HVA generated by the **originating desks**, denoted by H_j^c for desk $j \neq X$ and counterparty c . Consistently with the modelling assumptions, we attribute all counterparty credit and closeout effects to the XVA desk. As a consequence, the originating desks are only exposed to hedging frictions related to the underlying asset.

Proposition 4.5. (HVA dynamics for originating desks)

For originating desks $j \neq X$, the HVA components satisfy:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = f_j^S. \quad (61)$$

This equation describes the contribution to the HVA associated with a given originating desk j , driven by the cost of hedging its exposure to the underlying asset. The term f_j^S corresponds to the **gamma hedging cost** associated with desk j . This cost arises from the need to **continuously rebalance the hedge** by buying and selling the underlying asset S , in order to maintain the portfolio delta within a prescribed range.

Proof. We start from the expression (60) obtained previously, which describes the HVA dynamics aggregated across all desks j and counterparties c , including both the originating desks and the XVA desk. In order to analyse the contribution of the originating desks in isolation, we introduce a simplifying modelling assumption. We assume that all counterparty credit frictions f_c^C and closeout costs Ψ_c are entirely borne by the XVA desk.

Under this assumption, the originating desks are **only exposed to hedging frictions** arising from the **management of the underlying asset** S , and therefore only contribute through the asset-related friction terms f_j^S . Mathematically, this means that for $j \neq X$, the right-hand side of (60) only retains the terms f_j^S , while the remaining terms are attributed to the XVA desk. This yields:

$$\sum_{c, j \neq X} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = \sum_{j \neq X} f_j^S. \quad (62)$$

Finally, since the sum over $j \neq X$ is simply an **aggregation of independent desk contributions**, we can consider each originating desk separately. For a fixed desk $j \neq X$, we obtain:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = f_j^S. \quad (63)$$

□

Next, we derive an explicit representation of the HVA components associated with the originating desks.

Proposition 4.6. (Feynman–Kac representation of the HVA dynamics for originating desks)

The solution of (69) is given by:

$$H_j^c(t, S) = -\mathbb{E}_t^{\mathbb{Q}} \left[\int_t^T \exp \left(- \int_t^u (r_F + m_j^c \lambda_c) ds \right) g_j^c(u, S_u) du \right]. \quad (64)$$

Proof. In order to obtain a more explicit expression, we need to specify the **form of the friction term**. Following the approach introduced in [Burnett \(2021\)](#), we consider a simple model in which the hedging strategy is adjusted only when the **portfolio delta exceeds** a given **threshold**. In this setting, the resulting **friction** can be approximated as:

$$f_j^S = \left(\frac{\sigma \Gamma_j}{D_j} \right)^2 \psi_S(D_j), \quad (65)$$

where Γ_j is the **aggregate gamma of desk j** , D_j is the **rebalancing threshold**, and $\psi_S(\cdot)$ represents the **cost of trading a given quantity of the underlying asset**.

Next, we allocate the aggregate gamma of desk j across counterparties. More precisely, we write:

$$\Gamma_j = \sum_c \gamma_j^c, \quad (66)$$

where γ_j^c denotes the **contribution of counterparty c to the total gamma of desk j** .

Using this decomposition, the **asset-related friction term** f_j^S can be split across counterparties:

$$f_j^S = \sum_c f_j^{S,c}, \quad (67)$$

with

$$f_j^{S,c} = \left(\frac{\sigma}{D_j} \right)^2 \psi_S(D_j) \Gamma_j \gamma_j^c. \quad (68)$$

This reflects the idea that the **friction generated by gamma hedging** is **proportional** not only to the **total gamma of the desk**, but also to the **share of that gamma attributable to counterparty c** .

Substituting the decomposition of f_j^S in equation (63) and considering each counterparty contribution separately, we obtain, for each c ,

$$(\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c)) H_j^c = \left(\frac{\sigma}{D_j} \right)^2 \psi_S(D_j) \Gamma_j \gamma_j^c. \quad (69)$$

At maturity, the **hedging valuation adjustment vanishes**, so the terminal condition is:

$$H_j^c(T, S) = 0. \quad (70)$$

We can see that equation (69) is a **linear parabolic PDE** with source term:

$$g_j^c(t, S) = \left(\frac{\sigma}{D_j} \right)^2 \psi_S(D_j) \Gamma_j \gamma_j^c, \quad (71)$$

and discount rate:

$$k_j^c = r_F + m_j^c \lambda_c. \quad (72)$$

Therefore, by the *Feynman–Kac theorem*, its solution can be written as:

$$H_j^c(t, S) = -\mathbb{E}_t^{\mathbb{Q}} \left[\int_t^T \exp \left(- \int_t^u (r_F + m_j^c \lambda_c) ds \right) g_j^c(u, S_u) du \right]. \quad (73)$$

Equivalently, this can be rewritten as:

$$H_j^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_j^c \lambda_c}(t, u) \sigma^2 \frac{\psi_S(D_j)}{D_j^2} \Gamma_j \gamma_j^c \right] du, \quad (74)$$

where

$$D_{r_F + m_j^c \lambda_c}(t, u) = \exp \left(- \int_t^u (r_F + m_j^c \lambda_c) ds \right). \quad (75)$$

This expression shows that the HVA of the originating desks is primarily driven by **gamma exposure**. In particular, the **cost of hedging increases** with the **curvature of the portfolio**, reflecting the fact that **higher gamma implies more frequent rebalancing** and therefore **higher transaction costs**. $\square$

4.2 XVA desk HVA

Once we have characterised the HVA contributions of the originating desks, we now turn to the component of the adjustment that is borne by the XVA desk. By construction, this desk absorbs the hedging costs that are not attributed to the individual originating books.

Starting from the global HVA equation (60) and subtracting the contribution of the originating desks derived in the previous subsection, we obtain the dynamics of the XVA desk HVA.

Proposition 4.7. (HVA dynamics for the XVA desk)

The HVA components associated with the XVA desk satisfy:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_X^c = f_X^S + \sum_c f_c^\lambda + \sum_c \lambda_c \Psi_c, \quad (76)$$

where X denotes the XVA desk.

This expression shows that the HVA associated with the **XVA desk** is driven by **three distinct sources of cost**. The first term, f_X^S , corresponds to the **friction** generated by **hedging the desk's exposure to the underlying asset**. The second term, $\sum_c f_c^\lambda$, captures the **friction** arising from **hedging counterparty credit risk**. Finally, the term $\sum_c \lambda_c \Psi_c$ represents the **expected cost of unwinding hedging positions** when a **counterparty defaults**.

A useful feature of the expression (76) is that it separates naturally into a **continuous component** and a **default-related component**. For this reason, it is convenient to decompose the XVA desk HVA into two parts.

Definition 4.2. (Decomposition of XVA desk HVA)

The HVA of the XVA desk can be decomposed as:

$$H_X^c = H_{\text{Drag}}^c + H_{\text{Closeout}}^c, \quad (77)$$

where:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c = f_X^S + \sum_c f_c^\lambda, \quad (78)$$

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \sum_c \lambda_c \Psi_c. \quad (79)$$

The first term, H_{Drag}^c , corresponds to the part of the HVA generated by **day-to-day hedging frictions**, while the second term, H_{Closeout}^c , captures the **additional losses generated when hedging positions must be closed out at default**.

This decomposition is economically meaningful. The **drag** component reflects the **persistent and continuous cost of maintaining the hedge over time**, while the **closeout** component reflects **discrete losses triggered by counterparty default events**. Hence, unlike the originating desks, the XVA desk is exposed not only to standard rebalancing frictions but also to jump-related costs arising from the termination and restructuring of hedges at default. In this sense, the HVA of the XVA desk is richer than that of the originating desks. It combines market hedging frictions, credit hedging frictions, and closeout effects in a single valuation adjustment. This is precisely what makes the XVA desk central in the overall HVA framework.

4.2.1 Drag HVA

We focus on the **drag component** of the HVA associated with the XVA desk. This component captures the **continuous cost of dynamically maintaining the hedging strategy** over time.

Following the friction modelling approach introduced earlier, we assume that the hedging strategy is adjusted when exposures exceed certain thresholds. This leads to a friction cost that depends on the **gamma of the portfolio**. In this setting, two types of gamma contributions arise:

- (i) **Asset gamma**, denoted by Γ_{ss} , capturing sensitivity with respect to the underlying asset S ,
- (ii) **Credit cross-gamma**, denoted by Γ_{sc} , capturing the interaction between the underlying asset and counterparty credit risk.

We denote by γ_{ss}^c and γ_{sc}^c the contributions of counterparty c to these respective quantities. The **total friction cost** can then be expressed as a sum of asset and credit components.

Proposition 4.8. (PDE for drag HVA)

For each counterparty c , the drag component satisfies:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c = \left(\frac{\sigma \Gamma_{ss}}{D_s} \right)^2 \psi_s(D_s) + \left(\frac{\sigma \Gamma_{sc}}{D_c} \right)^2 \psi_c(D_c) \quad (80)$$

$$= \sum_c \left(\frac{\psi_s}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right). \quad (81)$$

Proof. Under a **delta-threshold rebalancing strategy** as done previously in equation (65), the friction costs can be expressed as:

$$f^S = \left(\frac{\sigma \Gamma_{ss}}{D_s} \right)^2 \psi_s(D_s), \quad f_c^\lambda = \left(\frac{\sigma \Gamma_{sc}}{D_c} \right)^2 \psi_c(D_c), \quad (82)$$

where D_s and D_c are the rebalancing thresholds, and ψ_s, ψ_c represent the transaction cost functions.

Next, assuming a **proportional allocation of gamma** across counterparties, we decompose the **asset friction** as:

$$f_S^c = \frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c, \quad (83)$$

while the **credit friction** remains:

$$f_c^\lambda = \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2. \quad (84)$$

Substituting these expressions into the drag HVA equation (78), we obtain the following PDE for each counterparty c :

$$(\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c = \frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2. \quad (85)$$

Summing over all counterparties yields:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c = \sum_c \left(\frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right), \quad (86)$$

which corresponds to the total drag contribution across the portfolio. $\square$

Recalling the statement of the *Feynman–Kac theorem*, it is possible to solve the PDE for H_{Drag}^c .

Proposition 4.9. (Feynman–Kac representation of drag HVA)

The drag component admits the representation:

$$H_{\text{Drag}}^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \left(\frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right) \right] du. \quad (87)$$

This representation shows that the drag HVA corresponds to the expected discounted value of future friction costs arising from continuous rebalancing of the hedging strategy.

Proof. In general, a PDE of the form:

$$\partial_t u(t, x) + \mu(t, x) \partial_x u(t, x) + \frac{1}{2} \sigma^2(t, x) \partial_{xx} u(t, x) - V(t, x) u(t, x) - f(t, x) = 0, \quad (88)$$

admits a representation as a discounted expectation. In our setting, we identify:

(i) the **source term** (friction costs):

$$f(t, S) = \frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2, \quad (89)$$

(ii) the **unknown function**:

$$u(t, S) = H_{\text{Drag}}^c, \quad (90)$$

(iii) the **discount rate**:

$$V(t, S) = r_F + m_X^c \lambda_c. \quad (91)$$

Applying the *Feynman–Kac theorem*, we obtain:

$$H_{\text{Drag}}^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \left(\frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right) \right] du. \quad (92)$$

□

This expression can be seen as a special case of the more **general multi-asset formulation**. In a multi-dimensional setting, the friction costs depend on the full covariance structure of the underlying risk factors and the portfolio Hessian. In that case, the drag HVA becomes:

$$H_{\text{Drag}}^c = - \sum_i \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \frac{\psi_i(D_i)}{D_i^2} \sum_{j,k} \rho_{jk} \sigma_j \sigma_k \Gamma_{ij} \gamma_{ik}^c \right] du. \quad (93)$$

Equivalently, the drag HVA can be written in compact form as:

$$H_{\text{Drag}}^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \text{Tr} (F \Gamma \Sigma \gamma^c) \right] du, \quad (94)$$

where F is a **diagonal matrix** containing the friction coefficients ψ_i/D_i^2 , Γ is the **Hessian matrix** of second-order sensitivities, Σ is the **covariance matrix** of the underlying risk factors, and γ^c represents the **contribution of counterparty c**.

4.2.2 Closeout HVA

We now turn to the **closeout component** of the HVA. This term captures the additional friction cost that arises when a counterparty defaults and part of the hedge must be unwound. In contrast with the drag component, which reflects the **continuous cost** of maintaining the hedge over time, the closeout HVA corresponds to a **discrete loss** triggered by the default event. More precisely, when counterparty c defaults, the XVA desk needs to unwind the **asset hedge** that was held against this counterparty, but not the **bond hedge**, since the latter is used directly to offset the default loss.

Proposition 4.10. (PDE for closeout HVA)

For each counterparty c , the closeout component of the HVA satisfies:

$$(\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \lambda_c \psi_s \left(\left| \partial_S V_c + \partial_S U_c + \sum_j m_j^c \partial_S H_j^c \right| \right). \quad (95)$$

Proof. We start from the closeout equation introduced in Definition 4.2:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \sum_c \lambda_c \Psi_c.$$

For a fixed counterparty c , this gives:

$$(\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \lambda_c \Psi_c. \quad (96)$$

Next, the **asset hedge** that must be **unwound** at the **default of counterparty c** is the **marginal hedge** associated with this counterparty. By definition, this position is:

$$\chi_c = -\partial_S \left(V_c + U_c + \sum_j m_j^c H_j^c \right). \quad (97)$$

Applying the **transaction cost function** ψ_s to the absolute size of this position, the unwind cost is:

$$\Psi_c = \psi_s(|\chi_c|). \quad (98)$$

Substituting this expression into (96) yields:

$$(\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \lambda_c \psi_s \left(\left| \partial_S V_c + \partial_S U_c + \sum_j m_j^c \partial_S H_j^c \right| \right).$$

□

As in the previous subsection, a closed-form representation can be obtained under a simplifying approximation. More precisely, we assume that the risk on the HVA itself is negligible compared to the total delta. This allows us to **neglect the contribution of $\partial_S H_j^c$** on the right-hand side of (95).

For notational convenience, we write:

$$\delta_c = \partial_S V_c. \quad (99)$$

Under this approximation, we obtain the following representation.

Proposition 4.11. (Feynman–Kac representation of closeout HVA)

Under the approximation,

$$\sum_j m_j^c \partial_S H_j^c \approx 0, \quad (100)$$

the closeout component admits the representation:

$$H_{\text{Closeout}}^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \lambda_c \psi_s(|\delta_c + \partial_S U_c|) \right] du. \quad (101)$$

If we neglect the XVA contribution inside the unwind delta, one obtains the approximation:

$$H_{\text{Closeout}}^c \approx - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \lambda_c \psi_s(|\delta_c|) \right] du. \quad (102)$$

Proof. Under approximation (100), equation (95) becomes:

$$(\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \lambda_c \psi_s(|\delta_c + \partial_S U_c|). \quad (103)$$

Since the **closeout HVA vanishes at maturity**, the terminal condition is:

$$H_{\text{Closeout}}^c(T, S) = 0. \quad (104)$$

Equation (103) is a **linear parabolic PDE** with source term:

$$g_c(t, S) = \lambda_c \psi_s(|\delta_c + \partial_S U_c|), \quad (105)$$

and discount rate:

$$k_c = r_F + m_X^c \lambda_c. \quad (106)$$

Applying the *Feynman–Kac theorem*, we obtain:

$$H_{\text{Closeout}}^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \lambda_c \psi_s(|\delta_c + \partial_S U_c|) \right] du. \quad (107)$$

Finally, a further approximation consists in neglecting the XVA contribution $\partial_S U_c$ inside the unwind delta, since it is of higher order than the underlying contribution δ_c . This gives:

$$H_{\text{Closeout}}^c \approx - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \lambda_c \psi_s(|\delta_c|) \right] du. \quad (108)$$

□

The total HVA associated with the XVA desk is obtained by summing the drag and closeout contributions. Combining (102) with the representation of H_{Drag}^c , we obtain:

$$H_X^c = - \int_t^T \mathbb{E}_t^{\mathbb{Q}} \left[D_{r_F + m_X^c \lambda_c}(t, u) \left(\frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2 + \lambda_c \psi_s(|\delta_c|) \right) \right] du. \quad (109)$$

This expression shows that the XVA desk HVA combines two distinct effects, the persistent cost of maintaining the hedge and the discrete loss associated with its termination.

Finally, the value of the coefficient m_X^c depends on the relative importance of these two components. In practice, it should typically lie between 1 and 2. When the drag contribution dominates, taking $m_X^c = 2$ is generally a reasonable approximation.

5 Numerical results

In this section, we present the **numerical results** associated with the HVA framework. We first describe the **implementation step by step**, in order to make the **Python code** clear and fully transparent, and then present and comment the results.

The section is divided into two parts. The first part is dedicated to the **replication of the figures and tables** presented in [Burnett and Williams \(2021\)](#). The second part focuses on a **simplified implementation of the HVA** based on market data, in order to illustrate how it varies with respect to key parameters.

5.1 Paper results replication

We consider a single counterparty holding a single 5-year, 1 billion foreign notional FX Forward. The setup uses a **Black–Scholes** formulation to price the CVA and then evaluates the XVA desk HVA via **Monte Carlo simulation** of equation (109). All interest rates are set to zero, which removes FVA from consideration and simplifies the discount factors. No collateral is assumed, and the linearity of the FX forward renders the originating desk’s HVA negligible, so we focus on the XVA desk HVA.

The full computation runs in approximately 3 minutes and 30 seconds (210.64s) on an NVIDIA Tesla T4 GPU, using $M = 100$ Monte Carlo paths and $N = 50$ time steps.

5.1.1 Parameters

The market and hedging parameters are summarised in Table 1.

Table 1: Parameters used in the numerical replication.

Category	Parameter	Value
Market	Foreign notional N	10^9
	FX spot S_0	0.7
	Log-normal volatility σ	7.5%
	Maturity T	5 years
Credit	Credit spread	100 bps
	Recovery rate R	40%
	Hazard rate $\lambda = \text{spread}/(1 - R)$	$0.01\bar{6}$
Asset hedging	Delta threshold D_s	50 000 dom/%
	Friction cost (spread to mid)	1 pip
Credit hedging	Delta threshold D_c	5 000 dom/bp
	Friction cost	5 bps
Other	$r_F = 0, \quad m_X^c = 2$	

5.1.2 CVA computation and Greeks

For what follows, we will need the following **Black–Scholes formulae**:

$$C^{BS}(S, K, r, \sigma, \tau) = S \Phi(d_1) - K e^{-r\tau} \Phi(d_2), \quad (110)$$

$$\Delta^{BS} = \frac{\partial C}{\partial S} = \Phi(d_1), \quad \Gamma^{BS} = \frac{\partial^2 C}{\partial S^2} = \frac{\phi(d_1)}{S \sigma \sqrt{\tau}}, \quad (111)$$

where:

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}, \quad (112)$$

$\Phi(\cdot)$ denotes the standard normal CDF and $\phi(\cdot)$ its density.

With zero interest rates, the positive exposure V_c^+ of an ATM FX forward coincides with a Black–Scholes call price. The CVA from the Feynman–Kac representation reduces to

$$\text{CVA}(S, t) = -\lambda(1 - R) N \int_0^{T-t} e^{-\lambda\tau} C^{BS}(S, K, \sigma, \tau) d\tau, \quad (113)$$

where C^{BS} denotes the Black–Scholes call price with zero rates. This integral is evaluated numerically using adaptive quadrature (`scipy.integrate.quad`).

The HVA requires the second-order sensitivities of the CVA. These are computed analytically by differentiating under the integral sign.

The **asset gamma** $\Gamma_{ss} = \partial_{SS}^2 \text{CVA}$ is obtained by replacing the call price with its Black–Scholes gamma:

$$\Gamma_{ss}(S, t) = -\lambda(1 - R) N \int_0^{T-t} e^{-\lambda\tau} \Gamma^{BS}(S, K, \sigma, \tau) d\tau. \quad (114)$$

The **asset–credit cross-gamma** $\Gamma_{sc} = \partial_{S\lambda}^2 \text{CVA}$ is obtained by differentiating the CVA with respect to both S and λ . The derivative with respect to λ brings down a factor $(1 - \lambda\tau)$ from the exponential and the prefactor, while the derivative with respect to S transforms the call into its delta:

$$\Gamma_{sc}(S, t) = -(1 - R) N \int_0^{T-t} e^{-\lambda\tau} (1 - \lambda\tau) \Delta^{BS}(S, K, \sigma, \tau) d\tau. \quad (115)$$

Both integrals are again evaluated via adaptive quadrature at each Monte Carlo node.

5.1.3 HVA via Monte Carlo

The total XVA desk HVA is given by equation (109). With $r_F = 0$ and $m_X^c = 2$, the discount factor simplifies to $D_{r_F + m_X^c \lambda}(0, u) = e^{-2\lambda u}$, and the formula becomes:

$$H_X^c = - \int_0^T \mathbb{E}^{\mathbb{Q}} \left[e^{-2\lambda u} \left(\underbrace{\frac{\psi_s}{D_s^2} \sigma^2 S^2 \Gamma_{ss}^2}_{\text{Asset HVA}} + \underbrace{\frac{\psi_c}{D_c^2} \sigma^2 S^2 \Gamma_{sc}^2}_{\text{Credit HVA}} + \underbrace{\lambda \psi_s(|\delta_c|)}_{\text{Closeout HVA}} \right) \right] du. \quad (116)$$

From expectation to Monte Carlo average. The risk-neutral expectation is approximated by the law of large numbers. We simulate M paths of the spot process under the risk-neutral measure (zero drift):

$$S_{t_{n+1}}^{(m)} = S_{t_n}^{(m)} \exp\left(-\frac{1}{2}\sigma^2\Delta t + \sigma\sqrt{\Delta t} Z_n^{(m)}\right), \quad Z_n^{(m)} \sim \mathcal{N}(0,1) \text{ i.i.d.}, \quad (117)$$

and replace the expectation with the empirical mean:

$$\mathbb{E}^{\mathbb{Q}}\left[e^{-2\lambda u} g(S_u, u)\right] \approx \frac{1}{M} \sum_{m=1}^M e^{-2\lambda u} g\left(S_u^{(m)}, u\right), \quad (118)$$

where $g(S, u)$ collects the three integrands in (116).

Trapezoidal discretisation of the time integral. We introduce a uniform grid $t_n = n\Delta t$, with $n = 0, 1, \dots, N$ and $\Delta t = T/N$. For each path m , define:

$$I_n^{(m)} = e^{-2\lambda t_n} g\left(S_{t_n}^{(m)}, t_n\right). \quad (119)$$

The trapezoidal rule gives:

$$\int_0^T e^{-2\lambda u} g(S_u^{(m)}, u) du \approx \Delta t \left(\frac{I_0^{(m)}}{2} + \sum_{n=1}^{N-1} I_n^{(m)} + \frac{I_N^{(m)}}{2} \right). \quad (120)$$

Fully discretised HVA. Combining the two approximations, the HVA estimator is:

$$H_X^c \approx -\frac{\Delta t}{M} \sum_{m=1}^M \left(\frac{I_0^{(m)}}{2} + \sum_{n=1}^{N-1} I_n^{(m)} + \frac{I_N^{(m)}}{2} \right), \quad (121)$$

where the integrand at each node is:

$$I_n^{(m)} = e^{-2\lambda t_n} \left[\frac{\psi_s(S_n^{(m)})}{D_s(S_n^{(m)})^2} \sigma^2 (S_n^{(m)})^2 \Gamma_{ss}(S_n^{(m)}, t_n)^2 + \frac{\psi_c}{D_c^2} \sigma^2 (S_n^{(m)})^2 \Gamma_{sc}(S_n^{(m)}, t_n)^2 + \lambda \psi_s(N) \right]. \quad (122)$$

The Monte Carlo standard error is estimated as:

$$\varepsilon = \frac{\hat{\sigma}_J}{\sqrt{M}}, \quad \hat{\sigma}_J^2 = \frac{1}{M-1} \sum_{m=1}^M \left(J^{(m)} - \bar{J} \right)^2, \quad (123)$$

where $J^{(m)}$ denotes the trapezoidal integral along path m and $\bar{J} = \frac{1}{M} \sum_m J^{(m)}$.

Translation of hedging parameters. An important step is the conversion of the paper's hedging thresholds into the **natural units**. The asset threshold $D_s = 50,000$ domestic units per percent means that the hedge is rebalanced when the P&L per 1% spot move exceeds this value. In natural (foreign) units, this corresponds to $D_s^{\text{nat}}(S) = D_s / (0.01 \cdot S)$, and the friction cost per rebalancing is $\psi_s(S) = 0.0001 \cdot D_s^{\text{nat}}(S)$ (one pip times the quantity traded). For credit, the threshold is $D_c = 5,000$ domestic per basis point, and the cost simplifies to $\psi_c = 5 \text{ bps} \times D_c = 25,000$ domestic units. The closeout term uses $\delta_c = N_{\text{foreign}}$ (constant for an FX forward), giving a constant integrand $\lambda \cdot 0.0001 \cdot N$.

5.1.4 Results

Table 2 compares our **day-zero estimates** with those reported in [Burnett and Williams \(2021\)](#). The discrepancies are attributable to the modest computational budget used in our implementation, which was chosen for practical runtime constraints. Despite this, the qualitative structure of the results is well reproduced: the credit HVA dominates the total adjustment, and the overall HVA represents a significant surcharge relative to the CVA, even if the latter remains much higher.

Table 2: Day-zero HVA numbers. Values are in domestic currency units.

Quantity	Value	Error	Paper
CVA	−1,484,293		−1,578,792
Closeout HVA	−7,676	±0	−5,783
Asset HVA	−4,032	±142	−3,018
Credit HVA	−138,276	±9,171	−442,447
Total HVA	−149,984	±9,172	−451,248
HVA / CVA	10.10%		28.58%

Figure 2 shows the **roll-down of the second-order CVA risks** over the lifetime of the trade. The left panel compares the expected asset gamma $\mathbb{E}[\Gamma_{ss}]$ with $\sqrt{\mathbb{E}[\Gamma_{ss}^2]}$, the latter being the quantity that drives the HVA through its quadratic dependence on gamma. The right panel displays the asset–credit cross-gamma, which **decays approximately linearly** over the life of the trade.

Figure 2: Decay profile of the second-order CVA risks over the lifetime of the trade. Left: expected asset gamma (green) compared with the square root of the expected squared gamma (blue). Right: asset–credit cross-gamma.

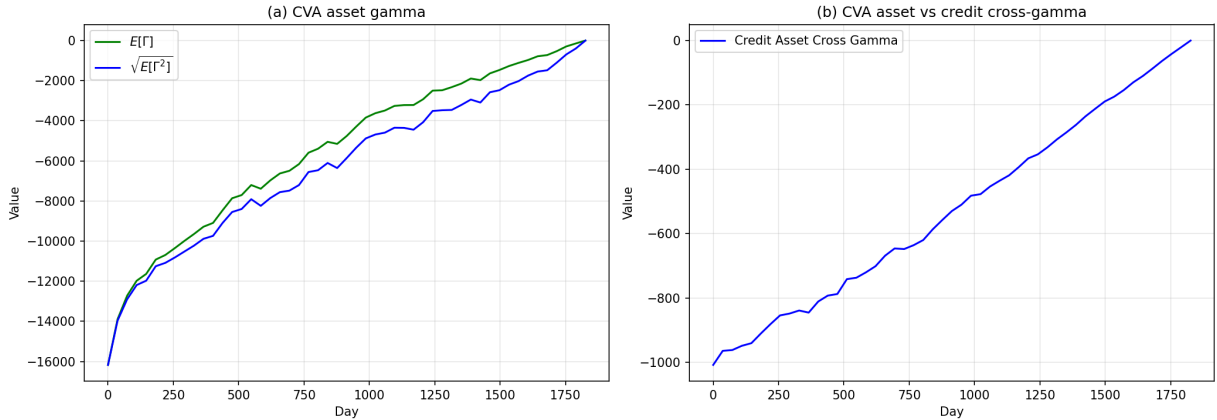
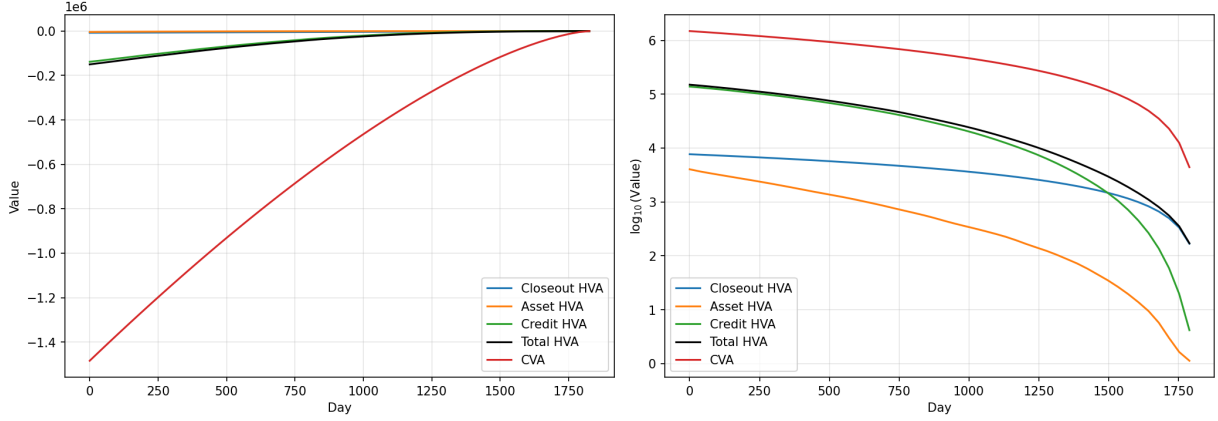


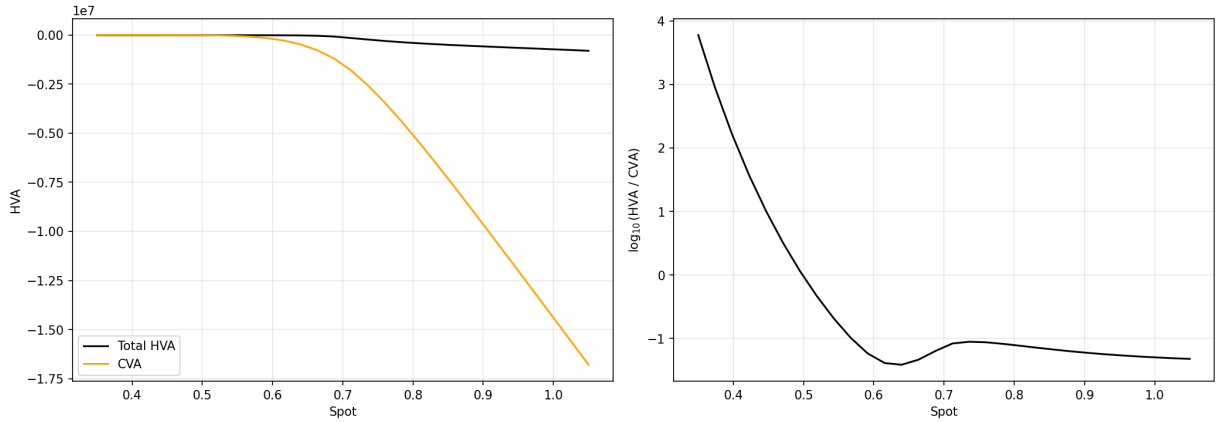
Figure 3 presents the **run-down profile of each HVA component alongside the CVA**. The left panel uses a linear scale, while the right panel shows the same quantities in log space (logarithms to base 10). As expected, **the credit component of the HVA dominates** over the asset and closeout components, reflecting the difference in liquidity between credit and FX markets. The total HVA line is almost indistinguishable from the credit HVA.

Figure 3: Run-down profile of each HVA component over time, plotted alongside the CVA. Left: linear scale. Right: logarithmic scale ($\log_{10}$).



Finally, Figure 4 shows the **variation of HVA and CVA** with the **spot level**, i.e. the moneyness of the underlying position. The left panel displays the absolute values, while the right panel shows the logarithm of the ratio HVA/CVA. For deeply out-of-the-money positions (low spot values), **the HVA can significantly exceed the CVA**, although both quantities are relatively small in absolute terms in that region.

Figure 4: Variation of HVA and CVA with spot level. Left: absolute values. Right: $\log_{10}(\text{HVA}/\text{CVA})$.



5.2 Application to real data

In this second part, we complement the paper replication with a simplified HVA calculation applied to real market data. The goal here is not to reproduce the full Monte Carlo machinery of equation (109), but rather to illustrate the **qualitative behaviour** of the HVA and its sensitivity to key parameters. For this reason, we adopt a first-order approximation of the drag HVA based on the friction rate formula alone, which avoids the computational cost of simulating CVA Greeks along stochastic paths.

5.2.1 Data

We select the **West Texas Intermediate** (WTI) crude oil spot price (DCOILWTIC0 from FRED) as our underlying asset due to its high liquidity and benchmark status for illustrating hedging-cost effects. We use three years of daily spot prices and compute log-returns:

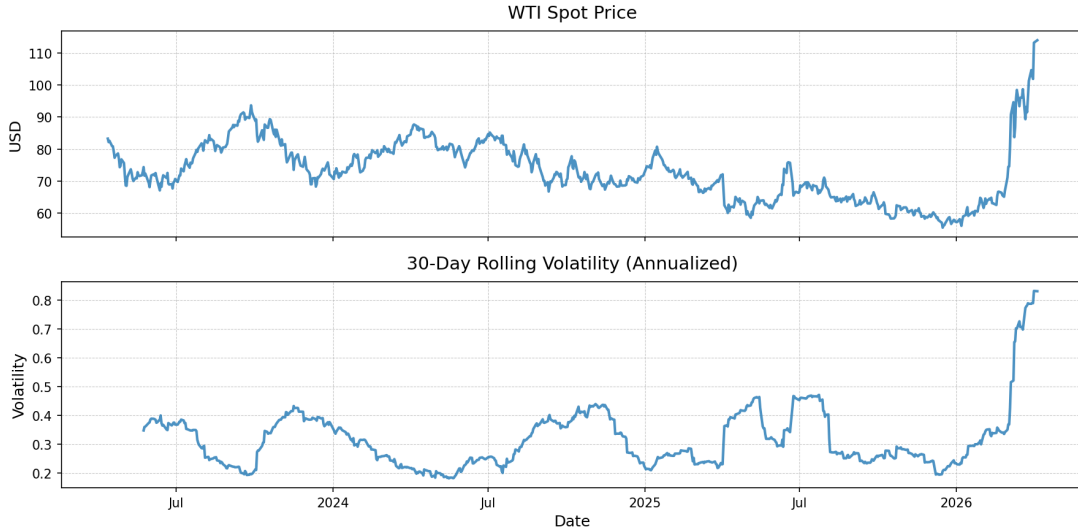
$$r_t = \ln \frac{S_t}{S_{t-1}}, \quad (124)$$

along with the 30-day rolling realised volatility, annualised by a factor of $\sqrt{252}$:

$$\sigma_{30}(t) = \sqrt{252} \text{Std}(r_{t-29}, \dots, r_t). \quad (125)$$

Over our sample, the average annualised volatility is $\bar{\sigma} \approx 35.25\%$. Figure 5 displays the **spot price** and the **rolling volatility**. The series exhibits typical commodity dynamics: sharp directional moves interspersed with calmer periods, and pronounced volatility clustering. There is clearly a spike in price and volatility following the outbreak of the war in Iran.

Figure 5: Top: daily WTI crude oil spot price. Bottom: 30-day rolling realised volatility (annualised).



5.2.2 Black-76 pricing and Greeks

Since the underlying reference is a commodity forward, the **Black-76** model is the natural pricing framework. Under this model, the forward price F_t follows a driftless log-normal process under the T -forward measure, and a European call with strike K and maturity T has price:

$$C^{76} = e^{-rT} (F_0 \Phi(d_1) - K \Phi(d_2)), \quad d_{1,2} = \frac{\ln(F_0/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad (126)$$

The associated Greeks are:

$$\Delta^{76} = e^{-rT} \Phi(d_1), \quad \Gamma^{76} = \frac{e^{-rT} \phi(d_1)}{F_0 \sigma \sqrt{T}}, \quad (127)$$

We calibrate the model with $F_0 = S_{\text{last}}$ (the most recent spot), $K = F_0$ (ATM), $T = 0.5$ years, $r = 3.6\%$ (around the 3 months US treasury bill yield in this period), and $\sigma = \bar{\sigma} \approx 35.25\%$ (these are very rough approximations, but still useful in order to keep the analysis as simple as possible). This yields an ATM call price of approximately 11.11 USD, with $\Delta^{76} \approx 0.540$ and $\Gamma^{76} \approx 0.014$.

5.2.3 Simplified HVA estimate

In this single-asset, single-counterparty setting without default risk, the closeout HVA vanishes and the credit cross-gamma term is absent. Then, the HVA reduces to the expected present value of the continuous drag cost. Adopting the friction rate formula from equation (65), the daily drag cost is:

$$f_S = \left(\frac{\sigma \Gamma}{D} \right)^2 \psi, \quad (128)$$

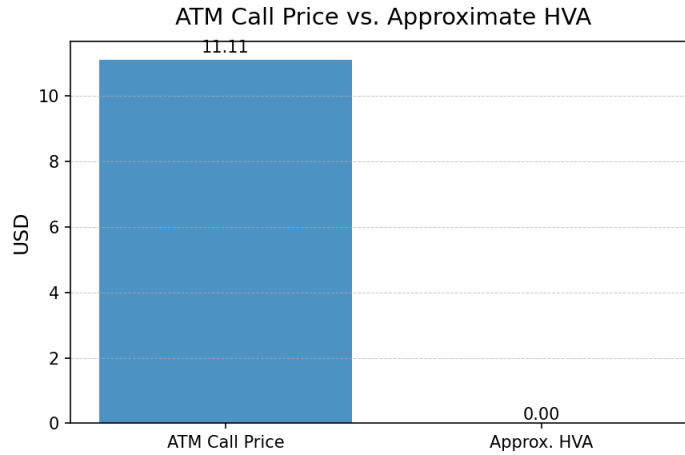
and the approximate HVA over T years is obtained by simple time integration:

$$\text{HVA}_{\text{approx}} = f_S \times 252 T. \quad (129)$$

This is clearly a **rough first-order estimate**: it freezes the gamma at its initial ATM value, ignores discounting and stochastic evolution of the spot, and omits all credit-related terms. However, it is enough to capture the correct functional dependencies and illustrate the qualitative behaviour of the adjustment.

For a **baseline calibration** with hedging tolerance $D = 1000$ and per-unit transaction cost $\psi = 0.01$ USD, the resulting **HVA is negligible** compared to the option price. Figure 6 confirms this: the HVA bar is imperceptible next to the ATM call price. This is expected in a liquid futures market with tight spreads and moderate gamma.

Figure 6: Comparison of the six-month ATM call price (≈ 11.11 USD) and the approximate HVA (≈ 0) under baseline parameters ($D = 1000$, $\psi = 0.01$).



5.2.4 Sensitivity analysis

Although the HVA is **negligible** under baseline conditions, the drag-cost formula predicts clear functional dependencies: quadratic in $1/D$, linear in ψ , quadratic in σ , and linear in T . We verify each of these through parameter sweeps.

Figure 7 presents two complementary views. The left panel shows the **HVA as a function** of both the **hedging tolerance D** and the **transaction cost ψ** : for small D (tight reheding) and large ψ (wide spreads), the adjustment grows from negligible to economically meaningful. The right panel displays a **3D surface of HVA** as a function of D and volatility σ (with $\psi = 10$ USD fixed), confirming that also tight reheding and large volatility imply large HVA.

Figure 7: Left: heatmap of approximate HVA as a function of hedging tolerance D and transaction cost ψ . Right: 3D surface of HVA as a function of D and volatility σ (with $\psi = 10$ USD).

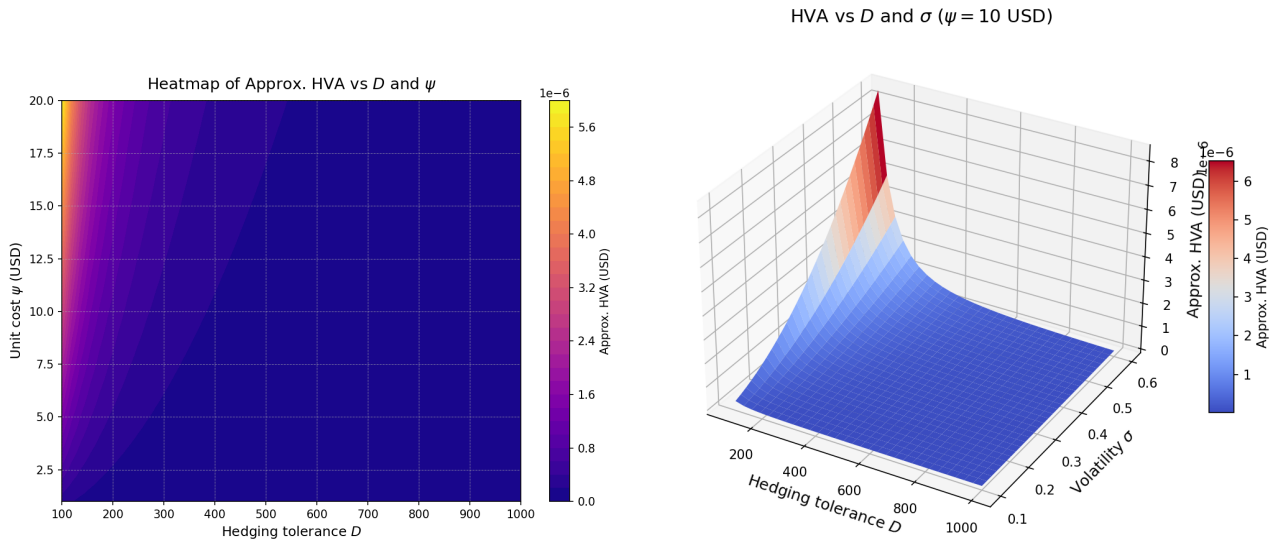
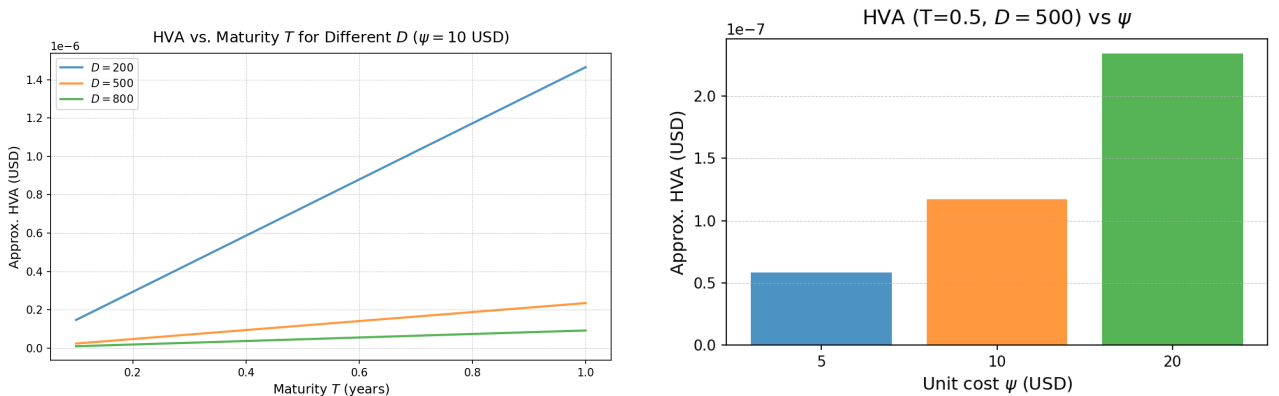


Figure 8 further illustrates the **remaining dependencies**. The left panel shows the **approximately linear growth** of HVA with maturity T for different values of D (with $\psi = 10$ USD), consistent with integrating a constant daily drag over the life of the option. The right panel confirms the **direct proportionality** $\text{HVA} \propto \psi$ at fixed $D = 500$ and $T = 0.5$.

Figure 8: Left: approximate HVA versus maturity T for different hedging tolerances D (with $\psi = 10$ USD). Right: approximate HVA versus transaction cost ψ (with $D = 500$, $T = 0.5$).



Overall, the sensitivity analysis recovers exactly the functional dependencies predicted by the drag-cost formula, validating the theoretical framework even in this simplified setting. From a practical standpoint, these results highlight that while HVA is negligible in tight-spread, moderate-gamma regimes, it can become **economically material under stressed conditions** characterised by higher volatility, wider bid–ask spreads, or more aggressive rehedgeing frequencies.

6 Conclusions

In this report, we have presented a detailed analysis of the **Hedging Valuation Adjustment** framework introduced in [Burnett \(2021\)](#) and extended in [Burnett and Williams \(2021\)](#), providing both a structured exposition of the **theoretical derivations** and a **numerical implementation** of the main results.

From a theoretical standpoint, we have shown how the HVA can be **rigorously incorporated** into the standard **XVA architecture**. Starting from a replication-based pricing framework augmented with hedging frictions, we derived a valuation equation in which the HVA appears as an **explicit adjustment** alongside CVA and FVA. The resulting decomposition into a **drag component**, capturing the continuous cost of delta rebalancing under bid–ask spreads and liquidity constraints, and a **close-out component**, reflecting the discrete loss incurred when hedges are unwound upon counterparty default, provides a transparent and economically interpretable structure. We have also seen that the HVA brings **peculiarities not present in other XVAs**, most notably the **non-linear dependence** on portfolio-level gamma, which prevents a simple additive allocation across counterparties and gives rise to the marginal multiplier m_j^c .

From a numerical standpoint, our replication of the paper’s results confirms the main **quantitative conclusions**, despite the use of a modest computational budget. In particular, the **credit component of HVA dominates** the total adjustment, driven by the asset–credit cross-gamma and the relatively high friction costs associated with trading credit instruments. Even in the simple case of a single FX forward, the total HVA represents a **non-negligible surcharge** relative to the CVA, suggesting that it cannot realistically be ignored in practice. Moreover, our **sensitivity analysis** based on real WTI crude oil data, while conducted under a simplified first-order approximation, recovers exactly the **functional dependencies** predicted by the theoretical framework: **quadratic sensitivity** to the inverse of the hedging tolerance, **linear dependence** on transaction costs, and **proportional growth** with maturity and volatility. These results highlight that the HVA, although negligible in liquid and tight-spread regimes, can become **economically material** under stressed market conditions or for portfolios with large unhedgeable cross-gamma exposures.

An important insight that emerges from both the theory and the numerics is that the bulk of real-world HVA is likely to originate from **cross-gammas** rather than from pure gamma risks. Pure gammas can often be neutralised by trading suitable options on the underlying, effectively eliminating their contribution to future hedging costs. Cross-gammas, however, represent risks that are **far harder to hedge** due to the lack of liquid hybrid instruments, leaving the trader with no better alternative than the delta-hedging strategy considered here. This is a problem that particularly affects **XVA desks**, given their inherently **cross-asset nature**.

Looking ahead, several directions could extend and refine the framework developed in this report. On the **modelling side**, relaxing the assumption of deterministic interest rates and credit spreads would allow for a richer interaction between market risk factors and hedging costs, potentially amplifying the HVA in realistic scenarios. The incorporation of **additional XVAs**, such as KVA, into the consistent framework would also be of interest, as capital requirements introduce yet another source of second-order cost that interacts with hedging frictions. On the **computational side**, the efficient estimation of HVA remains a challenge: the need to integrate over simulated XVA gammas along

Monte Carlo paths makes the calculation significantly more demanding than standard XVA computations, and **variance reduction techniques** or **regression-based approximations** could prove valuable. Finally, the analysis of **different hedging strategies** beyond the delta-threshold model, and in particular the determination of the expected P&L variance under each strategy, would provide a more complete picture of the trade-off between hedging cost and hedging quality that institutions face in practice.

Overall, the HVA formalism provides a **principled and quantitative answer** to a question that is already of daily relevance to traders: how to measure and account for the **systematic cost of hedging** in the presence of market frictions. While much work remains to be done on refining the methodology and scaling it to realistic portfolios, the basic message is clear: hedging costs represent a **very real economic burden** faced by financial institutions, and incorporating them explicitly into the valuation framework is both feasible and necessary.

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